

Miniaturization of popular reactions from the medicinal chemists' toolbox for ultrahigh-throughput experimentation

Received: 17 August 2022

Accepted: 22 May 2023

Published online: 22 June 2023



Nathan Gesmundo^{1,4}, Kevin Dykstra², James L. Douthwaite³, Yu-Ting Kao³, Ruheng Zhao³, Babak Mahjour³, Ron Ferguson², Spencer Dreher², Bérengère Sauvagnat^{1,5}, Josep Sauri^{1,6} & Tim Cernak^{1,3,7}✉

Miniaturization is a tactic employed in many technologies to accelerate discovery through parallelized data collection. The miniaturization of chemical synthesis to the limits of chemoanalytical and bioanalytical detection could accelerate drug discovery by increasing the amount of experimental data collected per milligram of material consumed. Here we demonstrate the miniaturization of popular reactions used in drug discovery such as reductive amination, *N*-alkylation, *N*-Boc (*tert*-butyloxycarbonyl) deprotection and Suzuki coupling for utilization in 1.2 μ l reaction droplets. Reaction methods were evolved to perform in high-boiling solvents at room temperature, enabling the diversification of precious starting materials, such as the complex natural product staurosporine.

Chemical space exploration in drug discovery generally requires access to many molecules with diverse physicochemical properties^{1–3}. The recent ability to computationally predict chemical properties and biochemical function with increasing accuracy has shifted the bottleneck of drug design to chemical synthesis⁴. In fact, chemical transformations determine product structure, so reactions themselves play an important role in influencing a compound's physicochemical properties⁵. Thus, indirectly, chemical transformations influence a compound's biological function. Recently, it has become increasingly possible to predict the outcome of chemical reactions with machine learning^{6,7}, although there are limited reaction data available to train such statistical models. For these reasons, technologies that accelerate and parallelize chemical synthesis could facilitate the systems-level navigation of an interconnected web of reaction condition, chemical structure, physicochemical property and biological function data⁸. Studies that intentionally optimize chemical reactions for miniaturization are lacking. To bridge this gap, we explore here the miniaturization of the most popular reactions used in drug discovery.

Inspired by Moore's Law of transistor miniaturization⁹, label-free reaction miniaturization has emerged as a technology with considerable promise to enable studies of the seemingly infinite chemical and reaction condition space. With reaction miniaturization^{10–12}, less starting material is needed to produce more reaction data, and the acquisition of each experimental data point requires less laboratory space, reaction vessel volume and produces less hazardous waste. This optimization increases the overall efficiency and 'greenness' of the exploration process¹¹. Based on miniaturization principles developed to enable combinatorial chemistry and high-throughput experimentation (HTE) on a 24, 96 or 384 reaction array, ultrahigh-throughput experimentation (ultraHTE)⁸ was developed to perform reactions in 1,536-well plates^{13–16}, giving increased freedom to survey reacting substrates and reaction conditions in concert. However, while reaction miniaturization may bolster the synthesis of bioactive compound libraries¹⁴ or the systems-level analysis of reaction performance⁸, few reactions are designed with 'above-the-arrow' reaction conditions that favour miniaturization. Meanwhile, a suite of trustworthy reactions—namely amide coupling, Suzuki coupling, heteroatom arylation such as

¹Merck & Co., Inc., Boston, MA, USA. ²Merck & Co., Inc., Kenilworth, NJ, USA. ³Department of Medicinal Chemistry, University of Michigan, Ann Arbor, MI, USA. ⁴Present address: AbbVie, North Chicago, IL, USA. ⁵Present address: Broad Institute of MIT and Harvard, Center for the Development of Therapeutics, Cambridge, MA, USA. ⁶Present address: Institut Químic de Sarrià, Barcelona, Spain. ⁷Present address: Department of Medicinal Chemistry, University of Michigan, Ann Arbor, MI, USA. ✉e-mail: tcernak@med.umich.edu

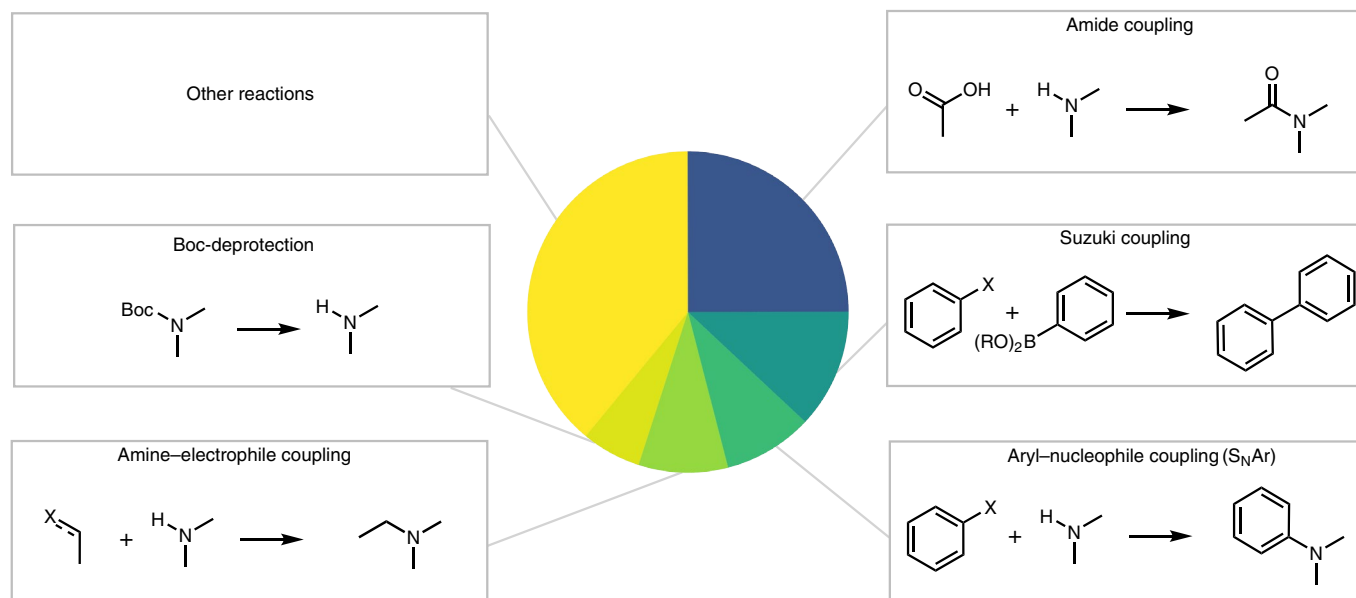


Fig. 1 | Popularity of common reactions in the synthesis of pharmaceuticals. Data collected from ref. 18.

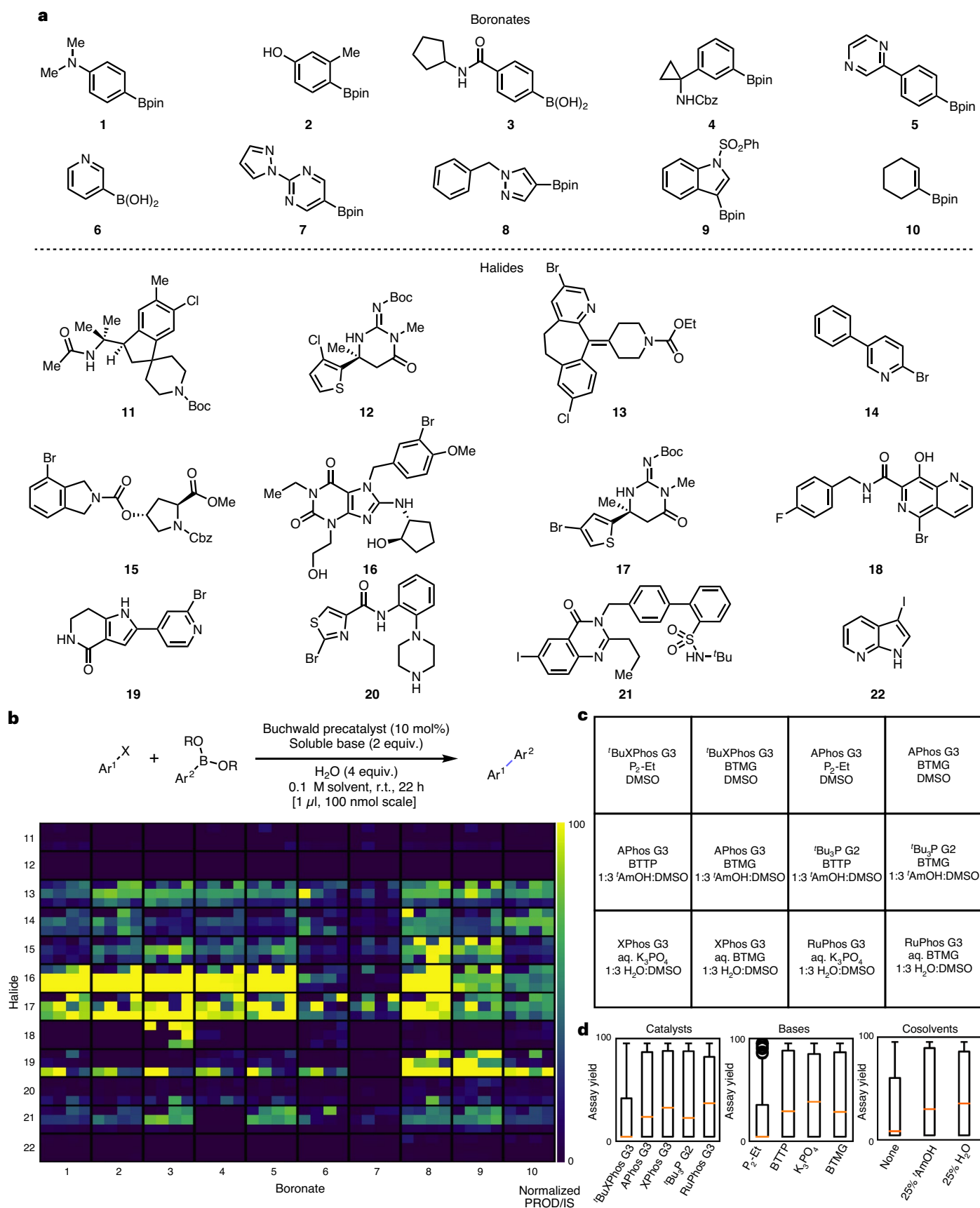
Buchwald–Hartwig coupling, heteroatom alkylation by reductive amination and *N*-Boc-deprotection—have emerged as reactions of choice for pharmaceutical chemical space exploration (Fig. 1)^{1,17–19}. These reactions are the workhorses of medicinal chemistry and allow many chemical analogues to be simultaneously prepared, either manually or automated in arrays of glass vials^{20,21}, in well plates^{14,16,19} or in flow^{22–24}. Developing on our earlier reports of miniaturized and automatable methods for amide coupling¹⁴ and Buchwald–Hartwig coupling^{13,14,16}, this current study explores miniaturized and automation-friendly variations of Suzuki coupling, reductive amination, *N*-alkylation and Boc-deprotection. Collectively, these reactions would cover two-thirds of transformations currently used by medicinal chemists (Fig. 1) and would vastly accelerate the exploration of chemical space, particularly through the direct-to-biology merger of chemical synthesis and bioassays¹⁴.

The robotic liquid handlers typically used in low-volume reagent dosing rely on involatile, homogeneous reagent stock solutions with low viscosity. Thus, high-throughput reaction methodology typically favours high-boiling solvents that fully dissolve all reaction components. In ultraHTE⁸, a typical reaction volume is ~1 µl, and solvents and reagents generally have a low enough vapour pressure that they can be handled without substantial evaporation during experiment setup. It is possible to handle volatile reagents or use volatile solvents by using liquid handlers that can rapidly deliver 384 reagent doses at a time¹⁵, or by employing a large dead volume of reagent stock solution, or running reactions at high dilution, typically in flow systems²⁵. However, increasing dead volume unnecessarily consumes precious starting materials, and reaction dilution is often detrimental to reaction kinetics. Therefore, the use of high-boiling solvents facilitates reaction miniaturization; it is likewise a remit of green chemistry^{26,27}. Solubility is also a key consideration in automated dosing. Techniques utilized in lower throughput reaction arrays to handle heterogeneous reagent mixtures, such as slurry loading¹⁰, present considerable engineering challenges during miniaturization. Therefore, selecting solvents that create homogeneous reagent mixtures ensures precise liquid-handling for reliable and consistent results. Recently, nanomolar solid dispensing has been demonstrated by coating reagents on glass beads²⁸. Miniaturization of homogeneous reactions that run to completion at room temperature is also desirable to avoid issues related to heat transfer, mass transfer and surface effects on insoluble particles,

making analysis and eventual scale-up of such reactions straightforward²⁹. Furthermore, reaction mixtures must be compatible with plastics that are routinely used as labware consumables on liquid-handling robots—typically high-density polyethylene, polypropylene and cyclic octane copolymer (COC). For these reasons, the initial development of nanoscale synthesis was predicated on fully homogeneous reactions that could be run at room temperature in high-boiling solvents like *N*-methyl-2-pyrrolidone (NMP), *N,N*-dimethylacetamide (DMAc) or dimethylsulfoxide (DMSO)^{14,16}. With this background we embarked on the miniaturization of popular reactions by identifying reaction conditions that would operate in an inert atmosphere glovebox at room temperature, in high-boiling solvents and in plastic labware. The timeframe needed to perform this workflow for a typical nanoscale study in a 1,536-well plate, from stock solution preparation through to data processing and visualization, takes approximately 1 week. Advancements in ultraHTE planning software and analytical methods are seeking to expedite this process even further^{30,31}.

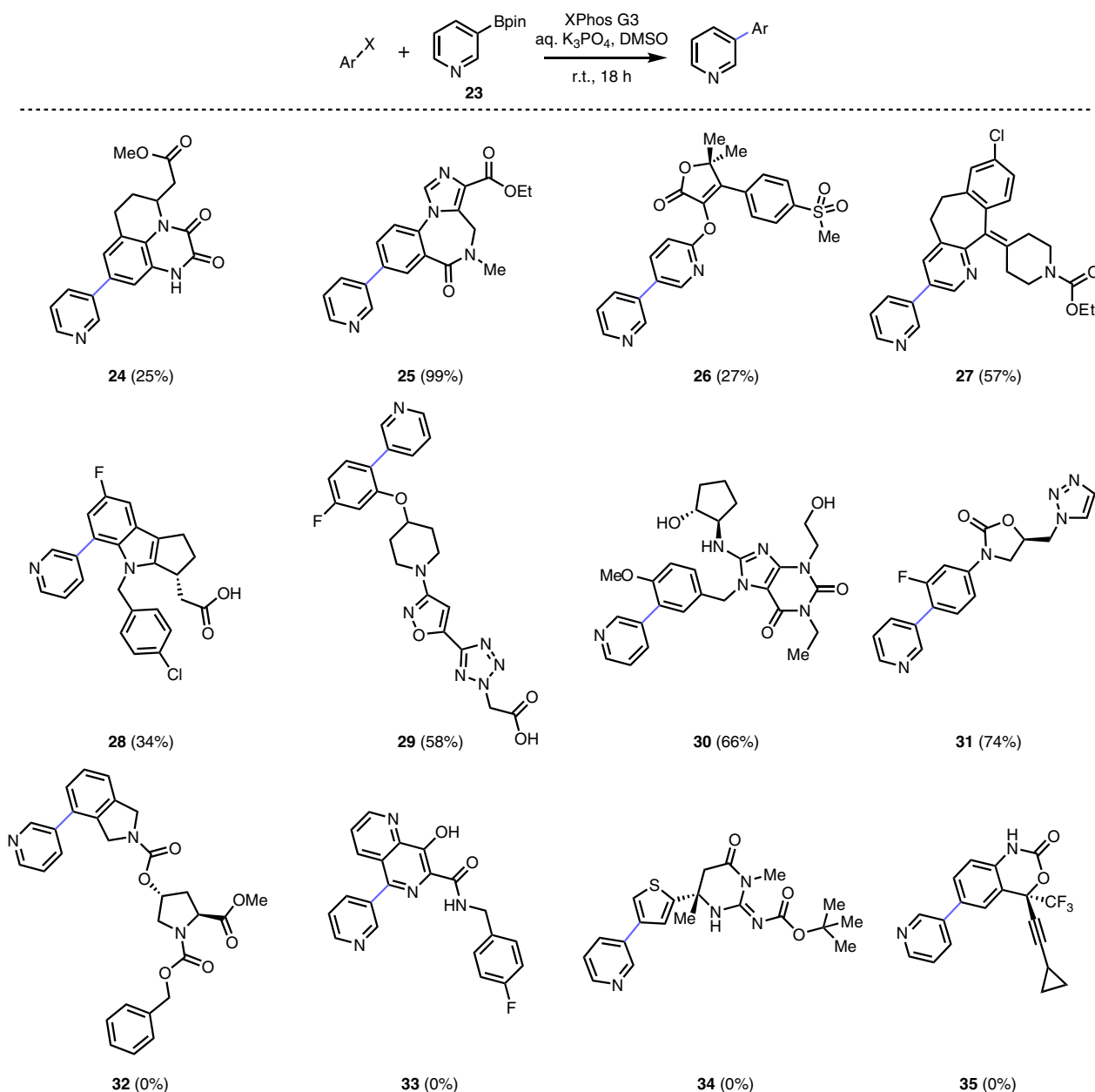
Given our earlier success in the miniaturization of Pd-catalysed C–N coupling using soluble superbases^{14,16}, we initiated our studies by targeting the Suzuki coupling—the second-most frequently used reaction in drug discovery. Our development of a Suzuki coupling protocol for ultraHTE was inspired by prior reports detailing the nanoscale use of Buchwald precatalysts with soluble organic bases^{16,32}. We were particularly interested in studying Suzuki coupling to heterocyclic bromides **19** and **20** for our reported kinase inhibitor campaign¹⁴, as the biaryl coupling of these substrates was unsuccessful under previously reported nanoscale Suzuki conditions¹⁶. The 2-dicyclohexylphosphino-2',4',6'-triisopropylbiphenyl (XPhos) G2 or G3 Buchwald precatalyst is known to readily couple aryl halides at room temperature; however, the preferred solvent for this reaction is tetrahydrofuran (THF)³³, which is too volatile to use in the nanoscale format and incompatible with the COC well plates we first employed. (Our first-generation studies in nanoscale synthesis used COC plastic plates which are readily degraded by THF. Subsequent reports have used polypropylene plastic plates which are resistant to many solvents.)

Additionally, aqueous K₃PO₄ is used as the base in these reactions. While mixtures of THF and aqueous K₃PO₄ form a biphasic mixture, we considered that aqueous K₃PO₄ may form a monophasic mixture or well-dispersed suspension in a polar aprotic solvent such as DMSO. We interrogated three solvent systems that contained

**Fig. 2 | Nanoscale ultraHTE Suzuki coupling reaction performance.**

a, Boronates (1–10) and halides (11–22) used in the study. **b**, Heatmap of conversion to product as determined by UPLC–MS relative to an internal standard at $\lambda = 254$ nm. **c**, Plate map. **d**, Box-plots comparing the effects of base,

cosolvent and catalyst on reaction performance (black circles indicate outlying points). Bpin, 4,4,5,5-tetramethyl-1,3,2-dioxaborolane; Cbz, benzyloxycarbonyl; ^tAm, *tert*-amyl; r.t., room temperature; PROD/IS, product/internal standard; aq., aqueous.

Table 1 | Coupling of Informer Library halides to 3-boronopyridine (23) using automation-friendly room temperature conditions

50 μmol scale. Yields are of purified products.

water or an alcohol as a cosolvent, given the importance of hydroxyl groups as activators in Suzuki coupling³⁴. Reactions using XPhos G3 or 2-dicyclohexylphosphino-2',6'-diisopropoxybiphenyl (RuPhos) G3 were dosed with an aqueous solution of K₃PO₄ or 2-*tert*-butyl-1,1,3,3-tetramethylguanidine (BTMG) and were run in 3:1 DMSO:water, whereas reactions with ^tBuXPhos G3, 4-(*N,N*-dimethylamino)phenyl di-*tert*-butyl phosphine (APhos) G3 or ^tBu₃P G2 were dosed with 1-ethyl-2,2,4,4,4-pentakis(dimethylamino)-2λ⁵,4λ⁵-catenadi(phosphazene) (P₂-Et), *tert*-butylimino-tri(pyrrolidino)phosphorane (BTTP) or BTMG, with four equivalents of water, and run in 100% DMSO or 25% ^tAmOH in DMSO. These 12 reaction conditions (Fig. 2c) were tested on eight previously reported Informer Halides (**11–13**, **15–18** and **21**)³⁵, two simpler heterocyclic halides (**14** and **22**) and bromides **19** and **20** (Fig. 2a)¹⁴. A corresponding array of boronates (**1–10**) included four diverse Informer Boronates^{35,36} and six boronates of interest

to our affinity ranking studies¹⁴. All 1,440 reactions (12 halides × 10 boronates × 12 reaction conditions) were performed in ultraHTE format in 1,536-well plates and their performance was evaluated by ultra-performance liquid chromatography–mass spectrometry (UPLC–MS) conversion to product compared with an internal standard (Fig. 2b).

By surveying reaction space in this systems-level format, several trends emerge. For instance, analysis of average reaction performance against various reaction parameters reveals that K₃PO₄ is a preferred base, with BTTP and BTMG having comparable but slightly lower performance, and the previously reported P₂-Et^{16,32} being the least successful in these Suzuki couplings (Fig. 2d). Additionally, anhydrous DMSO was considerably less productive than DMSO with ^tAmOH or water as cosolvent, which was anticipated³⁴. In terms of the phosphine ligand, ^tBuXPhos did not perform well in our study, whereas APPhos, ^tBu₃P, RuPhos or XPhos all gave acceptable catalysis. Given these data,

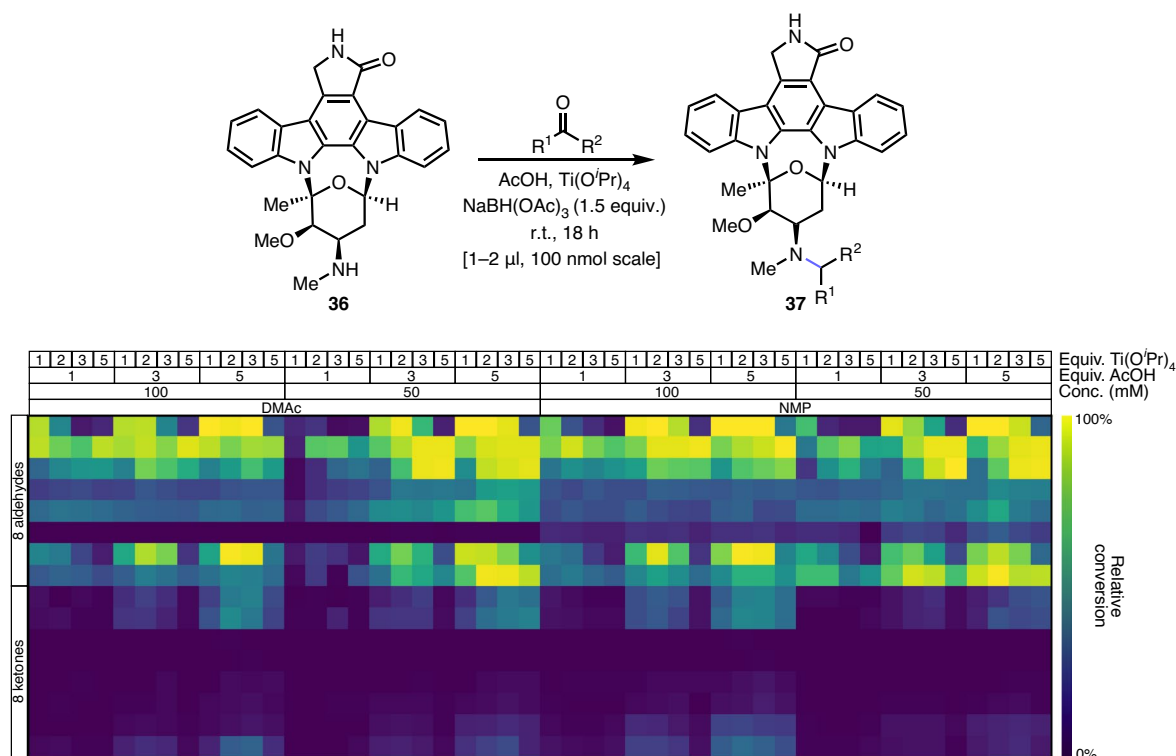


Fig. 3 | Nanoscale reaction optimization for the reductive amination of staurosporine (36) with eight aldehydes, eight ketones and 48 reaction conditions.

See Supplementary Section 5.1 for full list of aldehydes and ketones. Relative percentage conversion was determined by UPLC–MS at $\lambda = 254$ nm. Conc., concentration.

we chose XPhos G3 with aqueous K_3PO_4 in DMSO as the ideal conditions for subsequent coupling studies. To test reaction performance on a more traditional reaction scale, a variety of high-complexity halides from an Informer Library³⁵ were coupled to 3-boronopyridine (**23**) on a 50 μ mol scale at room temperature (Table 1). Of the 12 high-complexity halides employed, eight produced the coupled product (**24–31**) in 25% yield or greater using XPhos with aqueous K_3PO_4 in 3:1 DMSO:water. We believe substrate decomposition contributed to the failure of **32–35**. Nonetheless, given the complexity of the substrates explored, we believe that the use of XPhos G3 with aqueous K_3PO_4 in DMSO are viable room temperature reaction conditions for automated library synthesis in pharmaceutical chemical space.

Reductive amination is another of the most popular transformations used in medicinal chemistry^{18,19}. A common set of room temperature reaction conditions employs $NaBH(OAc)_3$ (ref. 37) and acetic acid in 1,2-dichloroethane or MeCN³⁸, while DMAc, another popular solvent³⁹, could be used for miniaturized synthesis. While reductive amination has been a popular choice for library synthesis in drug discovery on the microscale^{19,30,40}, we set our ambitions on the nanoscale diversification of staurosporine **36**, to showcase miniaturized analogue generation on a precious natural product. A preliminary study on the reductive amination of **36** with aldehydes and ketones revealed that $Ti(OiPr)_4$, a common additive^{41–43}, was key to achieving high conversion. We therefore investigated the reductive amination of **36** with eight aldehydes and eight ketones (see Supplementary Section 5.1 for structures) on the nanoscale across 48 reaction conditions (768 total reactions) to give analogues of type **37**, examining solvent, concentration and reagent loading simultaneously (Fig. 3). We found on average that between three and five equivalents of AcOH and two to three equivalents of $Ti(OiPr)_4$ gave the best conversion across the panel of substrates tested, particularly using NMP as the solvent, in which staurosporine was observed to be more soluble than in DMAc. Given that homogeneous reactions are much more amenable to miniaturization, we proceeded with NMP as the solvent for our next study.

Ketones showed low relative conversion across the range of reaction conditions tested with amine **36**.

Having explored many reaction conditions on few substrates, we next explored few reaction conditions on many substrates to build a library of staurosporine analogues through reductive amination with 48 diverse aldehydes (Fig. 4a; see Supplementary Section 5.2 for structures). We selected eight conditions from our previous screen for the study (two concentrations $\times$ two AcOH loadings $\times$ two $Ti(OiPr)_4$ loadings $\times$ 48 aldehydes) giving a total of 384 reactions. Of the 48 analogues targeted, 46 were successfully synthesized and detected by UPLC–MS. Selected reactions were repeated on a 50 μ mol scale and isolated yields of products (**45–50** and **51**) varied from 38% to 89%, with some of the lower yields attributed to observed instabilities of the purified products upon solvent removal (Fig. 4b). To further test the robustness of our reductive amination conditions (1.5 equiv. $NaBH(OAc)_3$, 2 equiv. $Ti(OiPr)_4$, 5 equiv. AcOH, 100 mM in NMP at room temperature), we completed the synthesis of analogues of three drug molecules on a larger 0.30 mmol scale (**51–53**) (Fig. 4c). It is likely that the free alcohol of **51** chelates the $Ti(OiPr)_4$ additive, and therefore a higher yield was obtained when this was omitted. Compounds **52** and **53** were successfully synthesized using the optimal general conditions for **36**.

A grand challenge of nanoscale chemistry operations will be the advent of multistep synthesis, which will enable deeper exploration of chemical space. We demonstrate here a simple entry to multistep nanoscale synthesis in the form of a selective *N*-alkylation and in situ Boc-deprotection. Base-promoted *N*-alkylations serve as a complementary approach to reductive amination. The *N*-alkylation of amides **54** and **55** with 12 primary, secondary, allyl and benzyl halides in the presence of four organic and organometallic bases was achieved on a 200 nmol scale (Fig. 5). Amides **54** and **55** were chosen as substrates as the products of these reactions are potent MK2 inhibitors^{44–46}. The regioselective alkylation of the less nucleophilic amide *N*-atom was made possible through Boc-protection of the more nucleophilic piperazine nitrogen.

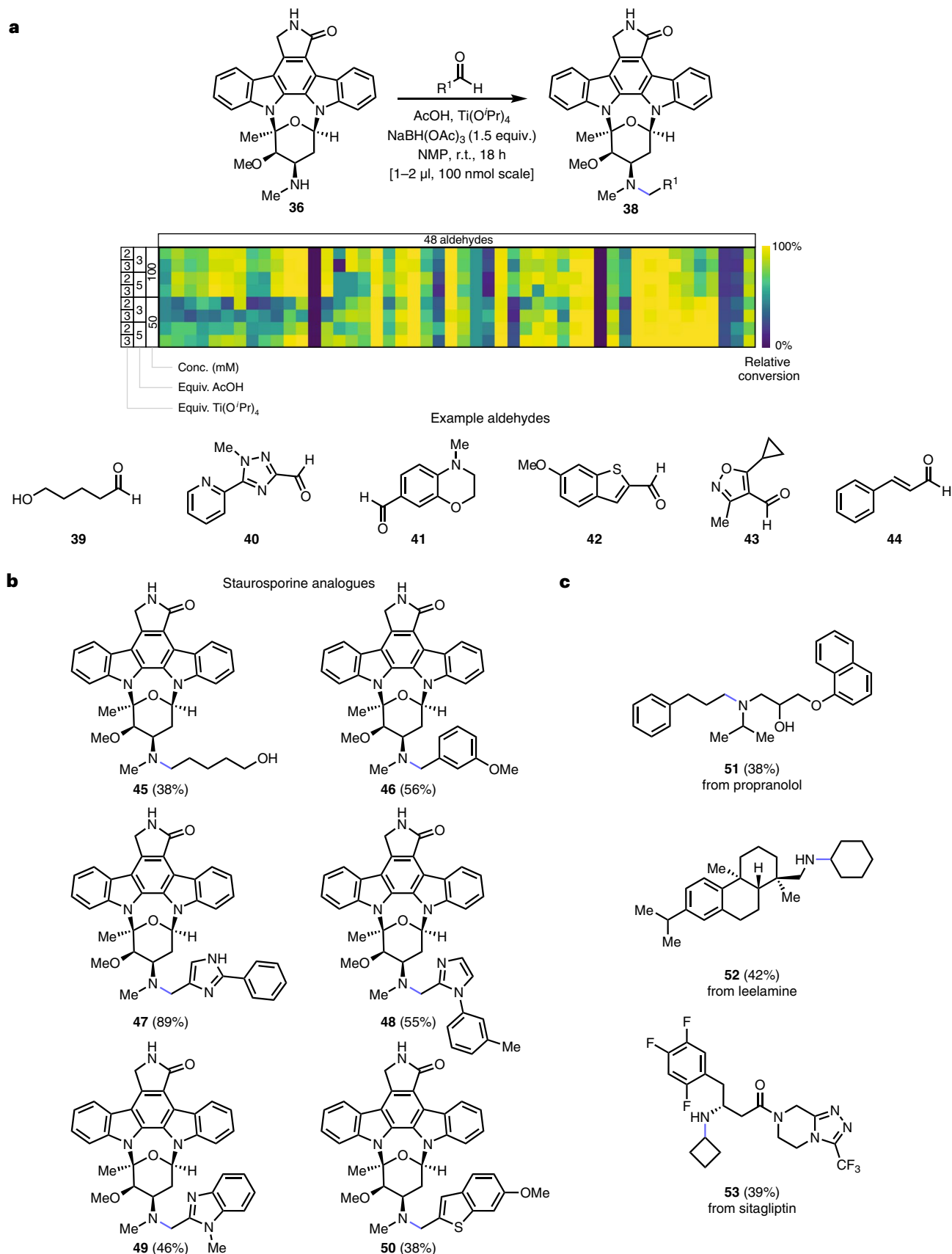


Fig. 4 | Nanoscale staurosporine analogue library synthesis via reductive amination. **a**, Heatmap of conversion to product for the reductive amination of staurosporine **36** with 48 aldehydes using eight reaction conditions. A representative example of aldehydes is shown (see Supplementary Section 5.2

for full list). Relative conversion was determined by UPLC–MS at $\lambda = 254$ nm. **b,c**, Selected reactions were repeated on 50 μ mol scale giving **45–50** and **51** (**b**) or 0.30 mmol scale giving **51–53** (**c**). Yields are of purified products.

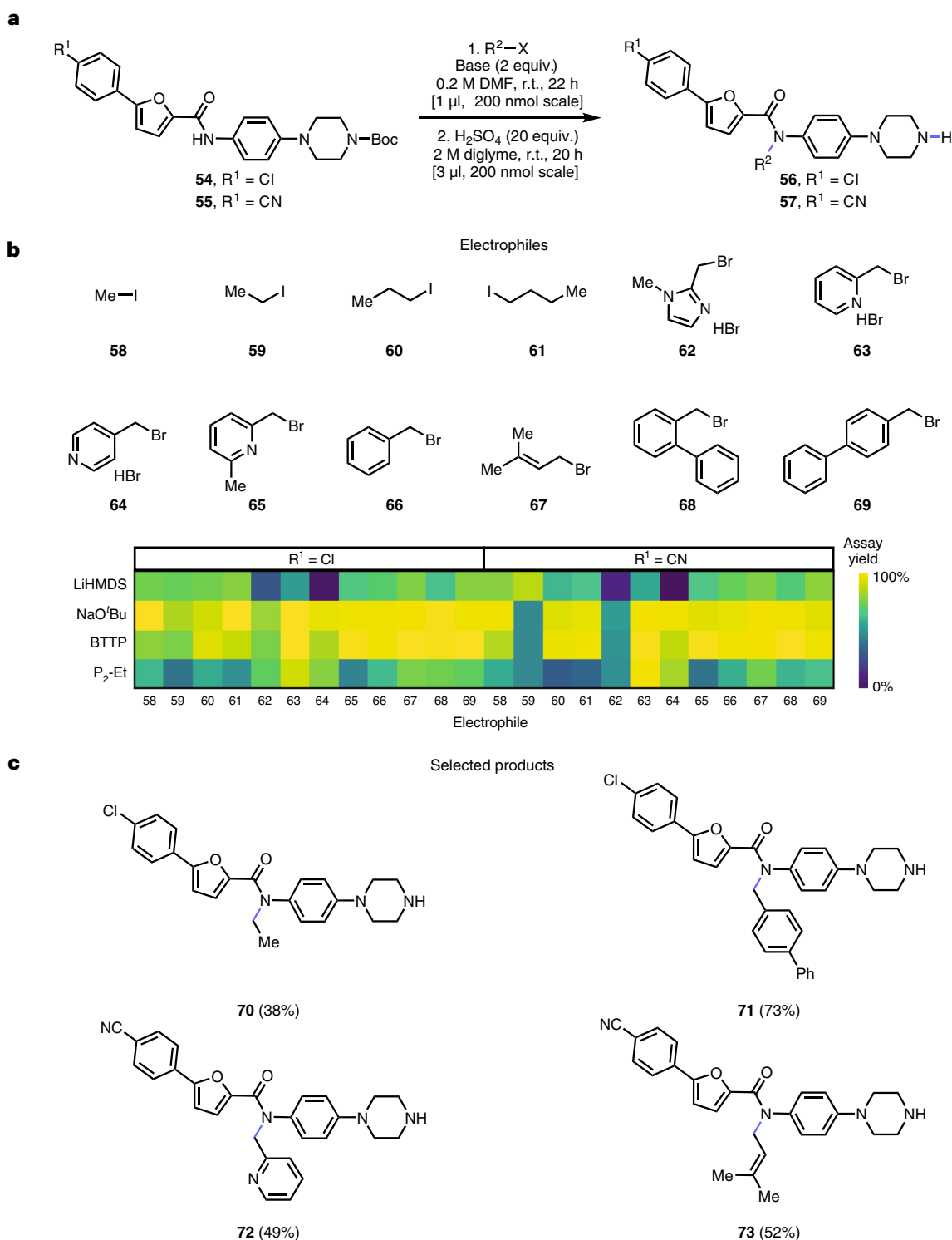


Fig. 5 | Two-step nanoscale synthesis of MK2-inhibitor analogues (56) and (57) via *N*-alkylation followed by Boc-deprotection. a, Electrophiles 58–69 were coupled to 54 and 55. b, Heatmap of reaction performance determined by UPLC–MS at $\lambda = 254$ nm relative to standard curves created from isolated products.

c, Selected examples 70–73 repeated on 40 μ mol scale (see Supplementary Section 6.5 for full scope). Yields are of purified products. LiHMDS, lithium bis(trimethylsilyl)amide.

The deprotection of Boc-groups is the fifth-most popular reaction used in drug discovery (Fig. 1)¹⁸, and thus represents another critical addition to the miniaturized synthetic chemistry toolbox. Following a survey of acids, we found that quenching reactions with sulfuric acid smoothly effected deprotection giving MK2-inhibitor analogues 56 and 57. While trifluoroacetic acid (TFA) was also a viable

acid, our reactions were performed in an inert atmosphere glovebox on a robotic system each containing sophisticated metal components, so we wanted to avoid the use of volatile acids like TFA and selected H₂SO₄ as a non-volatile alternative. (We caution that concentrated solutions of sulfuric acid will corrode components of the SPT mosquito robot so care should be used in handling these solutions.) The partnering of

base-promoted *N*-alkylation with in situ deprotection marked our first successful operation including more than one reaction operation in the ultraHTE format. From a screen of two-step reaction conditions, we identified NaO^tBu as being the most productive base for the first step across the electrophile library (**58–69**). Therefore, we applied optimal conditions of NaO^tBu in *N,N*-dimethylformamide (DMF) followed by H₂SO₄ in diglyme to the multistep synthesis of the 24 target compounds on the microscale, successfully isolating 21 products in yields of 12–73% for future biological testing (Supplementary section 6.5). Four such alkylated-deprotected compounds (**70–73**) are shown in Fig. 5c.

Chemical space exploration in drug discovery was one motivator in the development of the nanoscale synthesis platform¹⁶. It is therefore important that the favourite reactions from the medicinal chemist's toolbox are available in this miniaturized and automated format. In addition, modern synthetic developments such as C–H activation, should be continually translated to miniaturization⁴⁷. This report showcases key principles required to pursue reactions on the nanoscale effectively by performing homogeneous reactions, in high-boiling solvents, at room temperature upon a selection of biologically relevant substrates. While we find the conditions presented here to be general, it remains the case that many exact substrate pairs require boutique reaction conditions necessitating additional experimentation as new pockets of chemical space are explored. As miniaturized chemistry technology develops, there will be important chemical and engineering challenges to overcome so that chemists can incorporate the full breadth of modern chemical reactions and achieve further integration with bioassays¹⁴ and machine learning⁶. Ultimately, the link of transformations, reaction conditions, chemical structure, properties and biochemical function form an interconnected system of data that can be navigated through access to miniaturized reactions.

Methods

High-throughput experimentation

For nanoscale HTE all reactions were prepared at the 1 µl scale (0.1 M, 100 nmol) or 1.2 µl scale (0.1 M, 120 nmol) in a 1,536-well microplate using the SPT Labtech mosquito liquid-handling robot inside an N₂ atmosphere glovebox. In all cases, a 384-well source plate was created containing all required substrates, reagents, catalysts and building blocks (Greiner polypropylene deepwell 384-well plate, catalogue no. 784201). The SPT LabTech mosquito was used to dispense from the source plate into the 1,536-well reaction plate (Analytical Sales and Services, 20 µl polypropylene plate, catalogue no. 15020), mixing upon dispense, 3 × 750 nL. After dispensing, the reaction plate was sealed, centrifuged and allowed to sit at ambient temperature in the glovebox for 18–24 h. The next morning the reactions were quenched by the addition of 3 × 1 µl 0.5 M AcOH in DMSO (containing PhPh internal standard) to each reaction well, mixing upon dispense. After the nanoscale reactions were quenched, 1 µl of each quenched reaction was transferred, using the mosquito, from the 1,536-well reaction plate to a 384-well analytical plate (Greiner polypropylene deepwell 384-well plate, catalogue no. 784201) and diluted with 60 µl DMSO. The plate was sealed, shaken, then analysed by UPLC–MS. For screens where the total number of reactions was greater than 384, multiple 384-well analytical plates were prepared. For example, a 1,536 reaction screen requires four 384-well analytical plates. Images are shown in Supplementary Section 1.2 of key components of this miniaturized synthesis and screening platform. For a detailed step-by-step reaction setup guide to nanomole-scale HTE using the SPT Labtech mosquito, including images, see the preceding publication¹⁶. Reaction data were acquired with UPLC–MS, and processed using Virscidian Analytical Studio Software. See Supplementary Section 1.2 for more information on data acquisition and processing. Specific details for each experiment, such as source plate layout and order of mosquito protocol steps can be found in Supplementary Sections for each experiment.

General procedure for Suzuki coupling of 3-boronopyridine (**23**)

In a nitrogen-filled glovebox, pyridin-3-ylboronic acid **23** (0.060 mmol) and aryl informer (0.050 mmol) were combined with palladium Xphos G3 precatalyst (4.23 mg, 0.0050 mmol, 10 mol%) in anhydrous DMSO (0.1 M overall concentration) in 2 ml vials equipped with stir bars. To these mixtures, 50 µl of aq. K₃PO₄ (1.0 M aqueous solution, 0.05 mmol, 1 equiv.) was added and the mixtures were stirred at 23 °C overnight. The mixtures were quenched with 1% HOAc in DMSO (1.5 ml). The mixtures were then filtered through a filter plate (0.45 micron) and purified by reversed-phase high-performance liquid chromatography (RP–HPLC, Waters SunfireTM 5 micron RP C-18, 30 × 100 mm column, eluents: MeCN–H₂O containing 0.05% TFA).

General procedure for reductive amination on staurosporine (**36**)

Staurosporine **36** (23 mg, 0.05 mmol, 1 equiv.) was weighed into a 1-dram vial equipped with a magnetic stir bar. The vial was moved to a glovebox where NMP was added and the core dissolved. The desired aldehyde building block (1.2 equiv., 0.06 mmol, in NMP) was added to the vial, followed by acetic acid (5 equiv., 0.25 mmol, in NMP), and finally titanium tetraisopropoxide (2 equiv., 0.1 mmol, in NMP). The mixture was stirred at ambient temperature in the glovebox for 7 h. Finally, sodium triacetoxyborohydride (11.9 mg, 0.075 mmol, 1.5 equiv., added as stock solution in NMP) was added to the mixture, and the reaction stirred for 20 h. The next morning, aliquots were taken of all reactions and the progress was monitored by UPLC–MS. Upon completion, the reactions were quenched with sat. aq. NaHCO₃ (approx. 750 µl) then diluted with deionized water and dichloromethane. The organic phase was separated and the aqueous phase washed three times with dichloromethane. The combined organics were washed with brine, dried with MgSO₄ and concentrated. The crude residue was then taken in 2 ml DMSO, filtered and product submitted for purification by RP–HPLC.

General procedure for *N*-alkylation/Boc-deprotection

Tert-butyl 4-(4-(5-(4-chlorophenyl)furan-2-carboxamido)phenyl)piperazine-1-carboxylate **45** or *tert*-butyl 4-(4-(5-(4-cyanophenyl)furan-2-carboxamido)phenyl)piperazine-1-carboxylate **46** was weighed into a 1-dram vial equipped with a magnetic stir bar (19.25 mg or 18.89 mg, respectively, 40 µmol, 1 equiv.). The core was then dissolved in DMF and then NaO^tBu (80 µmol, 2 equiv.) was added as a slurry/solution in DMF. The reaction was stirred at room temperature for 30 min. Finally, the halide electrophile was added (50 µmol, 1.25 equiv.) and the alkylation was allowed to stir at room temperature overnight. The next morning, a 2 M stock solution of H₂SO₄ in diglyme was prepared and 400 µl was dispensed into the reaction vessel (800 µmol, 20 equiv.) and the reaction was stirred overnight at room temperature. The next morning, the reaction was quenched with 500 µl saturated aq. NaHCO₃. The reaction was diluted with DMSO, stirred and further quenched with carbonate base on a solid support. The reaction was filtered and submitted for purification by RP–HPLC. Products were returned as TFA salts unless otherwise noted. Many members of this library were previously disclosed through publication as MK2 inhibitors^{44,45}.

Data availability

The supporting information contains general information, experimental procedures, supplementary tables and images, and compound data and spectra. All raw high-throughput data can be accessed at <https://github.com/cernaklab/medchem-reaction-miniaturization/tree/main>.

References

1. Roughley, S. D. & Jordan, A. M. The medicinal chemist's toolbox: an analysis of reactions used in the pursuit of drug candidates. *J. Med. Chem.* **54**, 3451–3479 (2011).

2. Faller, B., Ottaviani, G., Ertl, P., Berellini, G. & Collis, A. Evolution of the physicochemical properties of marketed drugs: can history foretell the future? *Drug Discov. Today* **16**, 976–984 (2011).
3. Leeson, P. D. & Springthorpe, B. The influence of drug-like concepts on decision-making in medicinal chemistry. *Nat. Rev. Drug Discov.* **6**, 881–890 (2007).
4. Campos, K. R. et al. The importance of synthetic chemistry in the pharmaceutical industry. *Science* **363**, eaat0805 (2019).
5. Mahjour, B., Shen, Y., Liu, W. & Cernak, T. A map of the amine–carboxylic acid coupling system. *Nature* **580**, 71–75 (2020).
6. Shen, Y. et al. Automation and computer-assisted planning for chemical synthesis. *Nat. Rev. Methods Prim.* **1**, 23 (2021).
7. Shim, E. et al. Predicting reaction conditions from limited data through active transfer learning. *Chem. Sci.* **13**, 6655–6668 (2022).
8. Mahjour, B., Shen, Y. & Cernak, T. Ultrahigh-throughput experimentation for information-rich chemical synthesis. *Acc. Chem. Res.* **54**, 2337–2346 (2021).
9. Moore, G. M. Cramming more components onto integrated circuits. *Electronics* **38**, 114 (1965).
10. Krska, S. W., DiRocco, D. A., Dreher, S. D. & Shevlin, M. The evolution of chemical high-throughput experimentation to address challenging problems in pharmaceutical synthesis. *Acc. Chem. Res.* **50**, 2976–2985 (2017).
11. Wong, H. & Cernak, T. Reaction miniaturization in eco-friendly solvents. *Curr. Opin. Green Sustain. Chem.* **11**, 91–98 (2018).
12. Cernak, T. et al. Microscale high-throughput experimentation as an enabling technology in drug discovery: application in the discovery of (piperidinyl)pyridinyl-1H-benzimidazole diacylglycerol acyltransferase 1 inhibitors. *J. Med. Chem.* **60**, 3594–3605 (2017).
13. Uehling, M. R., King, R. P., Krska, S. W., Cernak, T. & Buchwald, S. L. Organic chemistry: pharmaceutical diversification via palladium oxidative addition complexes. *Science* **363**, 405–408 (2019).
14. Gesmundo, N. J. et al. Nanoscale synthesis and affinity ranking. *Nature* **557**, 228–232 (2018).
15. Lin, S. et al. Mapping the dark space of chemical reactions with extended nanomole synthesis and MALDI-TOF MS. *Science* **361**, eaar6236 (2018).
16. Santanilla, A. B. et al. Nanomole-scale high-throughput chemistry for the synthesis of complex molecules. *Science* **347**, 49–53 (2015).
17. Brown, D. G. & Boström, J. Analysis of past and present synthetic methodologies on medicinal chemistry: where have all the new reactions gone? *J. Med. Chem.* **59**, 4443–4458 (2016).
18. Boström, J., Brown, D. G., Young, R. J. & Keserü, G. M. Expanding the medicinal chemistry synthetic toolbox. *Nat. Rev. Drug Discov.* **17**, 709–727 (2018).
19. Dombrowski, A. W., Aguirre, A. L., Shrestha, A., Sarris, K. A. & Wang, Y. The chosen few: parallel library reaction methodologies for drug discovery. *J. Org. Chem.* **87**, 1880–1897 (2022).
20. Cooper, T. W. J., Campbell, I. B. & MacDonald, S. J. F. Factors determining the selection of organic reactions by medicinal chemists and the use of these reactions in arrays (small focused libraries). *Angew. Chem. Int. Ed.* **49**, 8082–8091 (2010).
21. Yu, Y. et al. Accelerating the discovery of DGAT1 inhibitors through the application of parallel medicinal chemistry (PMC). *Bioorganic Med. Chem. Lett.* **29**, 1380–1385 (2019).
22. Richardson, P. & Abdiaj, I. in *Flow Chemistry in Drug Discovery* (eds Alcazar, J. et al.) 421–479 (Springer, 2021).
23. Gioiello, A., Piccinno, A., Lozza, A. M. & Cerra, B. The medicinal chemistry in the era of machines and automation: recent advances in continuous flow technology. *J. Med. Chem.* **63**, 6624–6647 (2020).
24. Bogdan, A. R. & Organ, M. G. in *Flow Chemistry for the Synthesis of Heterocycles* (eds Sharma, U. K. & Van der Eycken, E. V.) 319–341 (Springer, 2018).
25. Perera, D. et al. A platform for automated nanomole-scale reaction screening and micromole-scale synthesis in flow. *Science* **359**, 429–434 (2018).
26. Koppmann, R. *Volatile Organic Compounds in the Atmosphere* (Blackwell Publishing Ltd, 2007).
27. Alder, C. M. et al. Updating and further expanding GSK’s solvent sustainability guide. *Green Chem.* **18**, 3879–3890 (2016).
28. Tu, N. P. et al. High-throughput reaction screening with nanomoles of solid reagents coated on glass beads. *Angew. Chem. Int. Ed.* **58**, 7987–7991 (2019).
29. Hendershot, D. C. & Sarafin, A. Safe chemical reaction scale up. *Chem. Health Saf.* **12**, 29–35 (2005).
30. Logsdon, D. L. et al. High-throughput screening of reductive amination reactions using desorption electrospray ionization mass spectrometry. *Org. Process Res. Dev.* **24**, 1647–1657 (2020).
31. Mahjour, B. et al. Rapid planning and analysis of high-throughput experiment arrays for reaction discovery. Preprint at *ChemRxiv* <https://doi.org/10.26434/chemrxiv-2022-hcnng> (2022).
32. Buitrago Santanilla, A., Christensen, M., Campeau, L. C., Davies, I. W. & Dreher, S. D. P2Et phosphazene: a mild, functional group tolerant base for soluble, room temperature Pd-catalyzed C–N, C–O, and C–C cross-coupling reactions. *Org. Lett.* **17**, 3370–3373 (2015).
33. Bruno, N. C., Tudge, M. T. & Buchwald, S. L. Design and preparation of new palladium precatalysts for C–C and C–N cross-coupling reactions. *Chem. Sci.* **4**, 916–920 (2013).
34. Thomas, A. A. & Denmark, S. E. Pre-transmetalation intermediates in the Suzuki–Miyaura reaction revealed: the missing link. *Science* **352**, 329–332 (2016).
35. Kutchukian, P. S. et al. Chemistry informer libraries: a chemoinformatics enabled approach to evaluate and advance synthetic methods. *Chem. Sci.* **7**, 2604–2613 (2016).
36. Dreher, S. D. & Krska, S. W. Chemistry informer libraries: conception, early experience, and role in the future of cheminformatics. *Acc. Chem. Res.* **54**, 1586–1596 (2021).
37. Abdel-Magid, A. F. & Mehrman, S. J. A review on the use of sodium triacetoxyborohydride in the reductive amination of ketones and aldehydes. *Org. Process Res. Dev.* **10**, 971–1031 (2006).
38. Abdel-Magid, A. F., Carson, K. G., Harris, B. D., Maryanoff, C. A. & Shah, R. D. Reductive amination of aldehydes and ketones with sodium triacetoxyborohydride. Studies on direct and indirect reductive amination procedures. *J. Org. Chem.* **61**, 3849–3862 (1996).
39. Afanasyev, O. I., Kuchuk, E., Usanov, D. L. & Chusov, D. Reductive amination in the synthesis of pharmaceuticals. *Chem. Rev.* **119**, 11857–11911 (2019).
40. Pels, K. & Kodadek, T. Solid-phase synthesis of diverse peptide tertiary amides by reductive amination. *ACS Comb. Sci.* **17**, 152–155 (2015).
41. Bhattacharyya, S. Titanium(IV) isopropoxide and sodium borohydride: a reagent of choice for reductive amination. *Tetrahedron Lett.* **35**, 2401–2404 (1994).
42. Mattson, R. J., Pham, K. M., Leuck, D. J. & Cowen, K. A. An improved method for reductive alkylation of amines using titanium(IV) isopropoxide and sodium cyanoborohydride. *J. Org. Chem.* **55**, 2552–2554 (1990).
43. Neidigh, K. A., Avery, M. A., Williamson, J. S. & Bhattacharyya, S. Facile preparation of *N*-methyl secondary amines by titanium(IV) isopropoxide-mediated reductive amination of carbonyl compounds. *J. Chem. Soc., Perkin Trans.* **116**, 2527–2531 (1998).
44. Huang, X. et al. Discovery and hit-to-lead optimization of non-ATP competitive MK2 (MAPKAPK2) inhibitors. *ACS Med. Chem. Lett.* **2**, 632–637 (2011).

45. Huang, X. et al. A three-step protocol for lead optimization: quick identification of key conformational features and functional groups in the SAR studies of non-ATP competitive MK2 (MAPKAPK2) inhibitors. *Bioorg. Med. Chem. Lett.* **22**, 65–70 (2012).
46. Fiore, M., Forli, S. & Manetti, F. Targeting mitogen-activated protein kinase-activated protein kinase 2 (MAPKAPK2, MK2): medicinal chemistry efforts to lead small molecule inhibitors to clinical trials. *J. Med. Chem.* **59**, 3609–3634 (2016).
47. Weis, E., Johansson, M., Korsgren, P., Martin-Matute, B. & Johansson, M. J. Merging directed C-H activations with high-throughput experimentation: development of iridium-catalyzed C-H aminations applicable to late-stage functionalization. *JACS Au* **2**, 906–916 (2022).

Acknowledgements

We thank S. Krska (Merck & Co., Inc.) for helpful discussions. This study was supported by the University of Michigan College of Pharmacy (start-up funds to T.C.), NIH (R01GM144471 to T.C.), the American Chemical Society MEDI Division (predoctoral fellowship to B.M.) and the National Defense Medical Center of Taiwan (fellowship to Y.-T.K.).

Author contributions

N.G., K.D., Y.-T.K., R.Z. and T.C. performed experiments. R.F. and S.D. assisted in preliminary experiments. B.S. and J.S. performed analysis of products. J.L.D. and B.M. and performed data analysis. N.G., J.L.D., B.M. and T.C. wrote the paper. All authors reviewed the data. T.C. supervised the work.

Competing interests

The majority of experimental work was performed at Merck & Co., Inc., Rahway, NJ, USA. The Cernak Lab at University of Michigan has received research funding or in-kind donations from MilliporeSigma, Relay Therapeutics, Janssen Therapeutics, Entos, Inc., SPT Labtech

and Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA. T.C. is a former employee of Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA, holds equity in Scorpion Therapeutics, and is a cofounder and equity holder of Entos, Inc. The other authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s44160-023-00351-1>.

Correspondence and requests for materials should be addressed to Tim Cernak.

Peer review information *Nature Synthesis* thanks Magnus Johansson, Steven Mennen and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editor: Peter Seavill, in collaboration with the *Nature Synthesis* team.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2023